weathering erosion & deposition stations

focus learning resources

Teacher's Notes

Thank you for purchasing my product!

I hope you are excited to start your students on these fun-filled stations. With some student practice, close teacher observation, and clear instructions from you, stations can be an effective and interactive teaching resource. To be the most effective, I recommend exposing the material to the students before you introduce the stations. This will allow the stations to reinforce and enhance the material your students have already learned.

The Lab Safety Stations from Focus Learning Resources© are designed to fit in a one-hour class period. You may need to adjust the time at each station depending on the length of your class period. The stations begin with a quick warm-up that students can complete while the teacher is taking attendance. This unit's warm-up is a foldable.

Before your students arrive, set up each *Station Card* on a different table or area of the room. You may use a lab station, a table, or a group of student desks as a station. Clearly label each station with large easy-to-spot numbers and the appropriate *Station Card*. Putting each *Station Card* in a page protector will keep it looking nice throughout all of your classes. As students arrive, have them take out a pen or pencil and then place all other personal items (i.e. their binders, backpacks, etc.) in a designated area of the classroom. In addition, give each student a *Student Workbook*, which they will carry with them as they travel to each station along with their writing utensil. You may also use the *Student Workbook* as their "ticket out the door" when class is over. Put 3-4 students in each station. If you have a larger class, you will start with 3-4 students at every station. However, if you have a smaller class, you do not have to fill all the stations with students as long as each group has the opportunity to visit each station.

If this is your first time to do stations with your students, take time the day before and immediately before the stations to thoroughly explain your expectations. Students should stay at each station until the teacher announces it is time to move. For this set of stations, the students will stay at each station 5-7 minutes; however, this is flexible. If a student group finishes that station and has leftover time, instruct students to continue working in areas of their *Student Workbook* they may not have finished or on their warm-up.

Happy Teaching!
Lauren Glenn
Focus Learning Resources©

Preparing for Lab Safety Stations

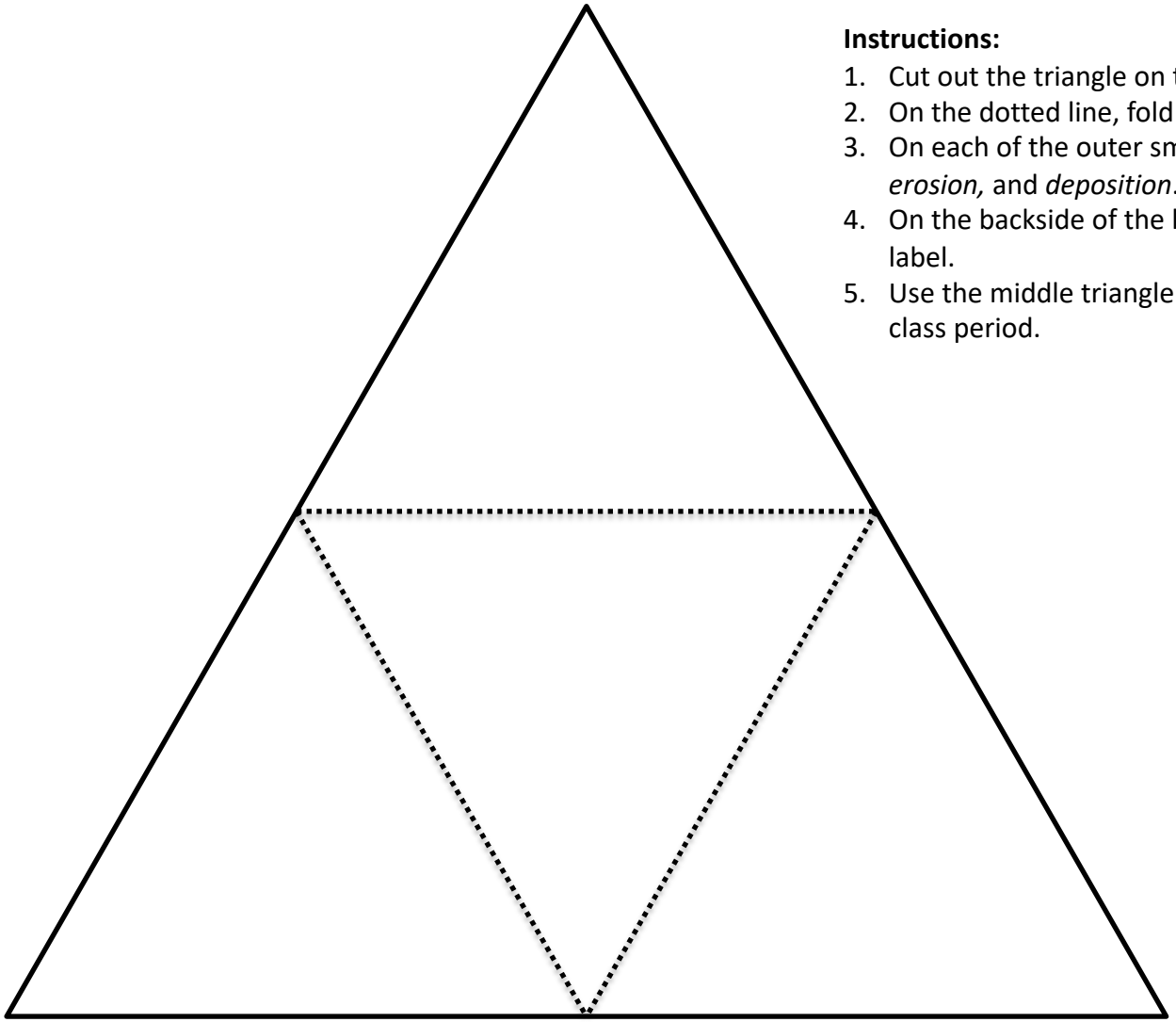
Materials:

- Copies of warm-up foldable (enough for each student). Students will need a personal pair of scissors and colored pencils for activity.
- Copies of *Student Workbook* (enough for each student)
- 1 copy of each *Station Card*
- 7 page protectors (one for each *Station Card*)
- Colored Pencils for Stations #1, 2, 5, 6, and 7
- Textbooks or other resources with examples of weathering, erosion, and deposition for Station #2
- Several books or articles on hurricanes for Station #3
- NERDS candy, enough for each student to have 2 NERDS each for Station #4
- Timer
- 2 small plastic tubs (shoebox size) for Statio
- 1 large plastic tub (as large as a student's desk, shallow)
- Sand
- Water
- Straws, enough for each group to have 1
- 2 spatulas
- 2 rolls of paper towels
- 5 pairs of goggles
- 2 trashcans
- 1 dustpan and duster
- 1 tennis ball
- Enough medium-sized rocks or building blocks to make a small, several inch high "mountain"

Teacher Preparation:

- Warm-up
  - Hand a copy of foldable to each student as they enter the classroom
- Station #1
  - Place *Station Card* and colored pencils at station
- Station #2
  - Place *Station Card* and colored pencils at station
- Station #3
  - Place *Station Card* at station
  - Place textbooks and WED books/resources at station
- Station #4
  - Place *Station Card,* candy, and timer at station
- Station #5
  - Place *Station Card* at station
  - Place enough sand to fill one side of a small plastic tub, and fill halfway with water
  - Place spatula, paper towels, and trashcan at station
- Station #6
  - Place enough sand to fill one side of a small plastic tub (no water for this tub)
  - Place tub, spatula, package of plastic straws, trashcan, dust pan and duster, and four pairs of goggles at station
- Station #7
  - Place large plastic tub and tennis ball at station
  - Place rocks or building blocks inside tub

WED Warm-up



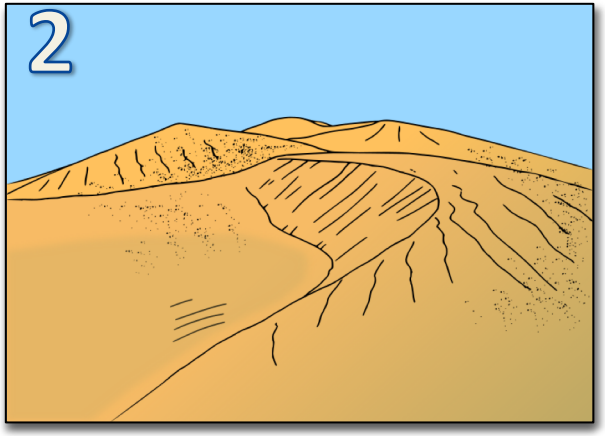
Instructions:

1. Cut out the triangle on the solid line.
2. On the dotted line, fold the smaller triangles inward.
3. On each of the outer small triangles, label: *weathering, erosion,* and *deposition*.
4. On the backside of the label, draw an example of the label.
5. Use the middle triangle to write your name and your class period.

Station #1: Identifying Weathering, Erosion, & Deposition

Materials:
Colored pencils

Instructions:
In your *Student Workbook*, note whether the following are pictures illustrate weathering, erosion, or deposition.
For #4, follow the directions in your *Student Workbook*.



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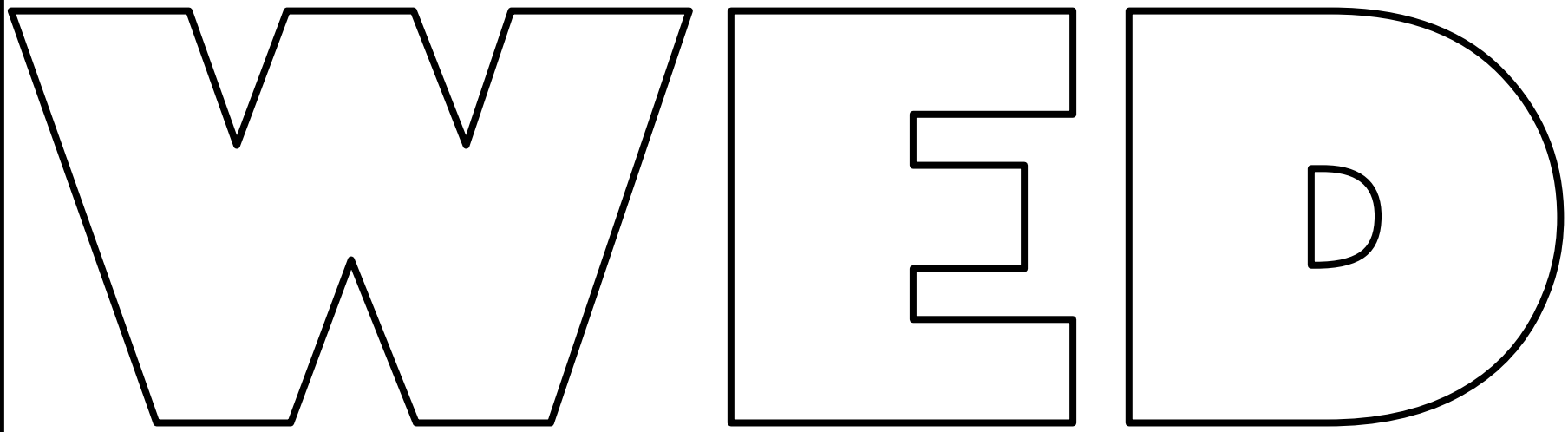
Station #2: Illustrating Weathering, Erosion, & Deposition

Materials:
Colored pencils

Instructions:
Using your textbook, notes, and other resources, find examples of weathering, erosion, and deposition. In your *Student* Workbook, draw an example of weathering in the W. In the *E*, draw an example of erosion. And in the *D*, draw an example of deposition.



Station #3: Researching Weathering, Erosion, & Deposition

Materials:
Resources on hurricanes, provided by teacher

Instructions:
Provided are several resources that give information on hurricanes. Use the information you find to describe how this event could cause weathering, erosion, and deposition. Write all your answers in complete sentences in your *Stations Workbook*.

Weathering

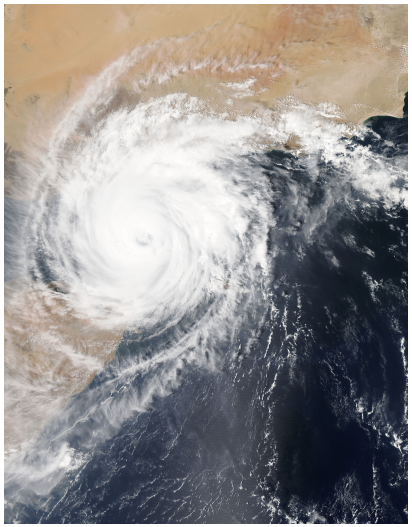
Give an example of the type of **weathering** that can occur during a **hurricane**

Erosion

Give an example of the type of **erosion** that can occur during a **hurricane**

Deposition

Give an example of the type of **deposition** that can occur during a **hurricane**



Station #4: Simulating Weathering, Erosion, & Deposition

Materials:
NERDS candy
Timer

Instructions:
Each one in your group gets two pieces of NERD candy. Your group will be simulating weathering, erosion, and deposition using NERDS. Follow the instructions below and record your observations in your *Stations Workbook*.

SIMULATION #1
Place one NERD in your mouth. DO NOT chew. Close your mouth and allow the NERD to dissolve. After 30 seconds, you may finish chewing the NERD. Record your observations and answer the questions in your *Student Workbook* for SIMULATION #1.

SIMULATION #2
Place one NERD in your mouth. This time chew the NERD and swallow. Record your observations and answer the questions in your *Student Workbook* for SIMULATION #2.

Station #5: Tropical Weathering, Erosion, & Deposition

Materials:
Small plastic tub full of water and sand (sand should be pushed all to one side of the plastic tub)
1 spatula
Paper towels
Trashcan

CAUTION: This is a very messy station! Follow the directions carefully and clean this station thoroughly before leaving for the next station.

Instructions:
1. Designate one person in your group to be the wave simulator. The wave simulator will ***gently*** rock the plastic tub back and forth simulating waves ebbing back and forth on the sandy shore. To simulate the waves should only take about 30 seconds.
2. Record your observations and answer the questions in your *Student Workbook*.
3. When finished, return the sand to the edge of the plastic bin and clean your area.

Station #6: Wind & Weathering, Erosion, & Deposition

Materials:
Small plastic tub with sand (sand should be pushed all to one side of the plastic tub)
1 spatula
Package of plastic straws
Trashcan
Dust pan and duster
Goggles

CAUTION: This is a very messy station! Follow the directions carefully and clean this station thoroughly before leaving for the next station.

Instructions:
1. Everyone in the group must wear goggles during this simulation.
2. Designate one person in your group to be the wind simulator.
3. Have the wind simulator ***gently*** blow through the straw horizontally (the same direction as the table, floor, or counter) at the sand. CAUTION: If you blow downward onto the sand, it will make a mess and the sand will blow into your face. The wind simulation should only take about 30 seconds.
4. Record your observations and answer the questions in your *Student Workbook*.
5. Return the sand to the edge of the plastic bin and clean your area.

Station #7: Gravity & Weathering, Erosion, & Deposition

Materials:
Large plastic tub
Tennis ball
Rocks or building blocks

Instructions:
1. Designate one person in your group to be the gravity simulator.
2. As a group build a small structure or "mountain" in the plastic tub out of the rocks or building blocks.
3. Have the gravity simulator ***gently drop*** (not throw) the tennis ball on top of the "mountain."
4. Record your observations and answer the questions in your *Student Workbook*.
5. If necessary, clean your station area.

Thank you!

Thank you for purchasing my product! If you and your students have enjoyed this resource, please follow me! I will continue to post more fun, interactive, and quality products like this one.

If you have any questions or comments, please feel free to email me personally: lauren.glenn10@gmail.com

Happy Teaching!
Lauren Glenn
Focus Learning Resources

Credits & Resources

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